

Program overview

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Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Applied Mathematics

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Bachelor TW 2023			
1e jaar BSc TW 2023			
AM1010	Mathematical Structures	6	
AM1020	Kaleidoscope	6	
AM1030	Linear Algebra 1	6	
AM1040	Analysis 1	6	
AM1050-A	Modelling A	3	
AM1050-B	Modelling B	3	
AM1060	Algebra 1	6	
AM1070	Analysis 2	6	
AM1080	Introduction to Probability Theory	6	
AM1090	Introduction to Programming	6	
keuzevakken 1e jaar TW 2023			
AM1510EE	Electricity and Magnetism	6	
AM1520TI	Algorithms and Data Structures	6	
AM1531TN	Mechanics and Theory of Relativity	6	
2e jaar BSc TW 2023			
AM2010	Linear Algebra 2	6	
AM2020	Optimization	6	
AM2030	Ordinary Differential Equations	6	
AM2040	Complex Function Theory	6	
AM2050-A	Modelling 2A	3	
AM2050-B	Modelling 2B	3	
AM2060	Numerical Methods 1	6	
AM2070	Partial Differential Equations	6	
AM2080	Introduction to Statistics	6	
AM2090	Real Analysis	6	
keuzevakken 2e jaar BSc TW 2023			
AM2510	Decision Theory	6	
AM2520-H	History of Mathematics	6	
AM2520-P	Philosophy of Mathematics	6	
AM2550	Advanced Statistics	6	
AM2560	Applied Algebra: Codes	6	
AM2570	Markov Processes	6	
AM2580	Mathematical Models in Biology	6	
3e jaar BSc TW 2023			
AM3000	Bachelor Project	18	
Minor (30 EC)			
keuzevakken 3e jaar BSc TW 2023			
AM3500	Mathematics Seminar	6	
AM3510	Mathematical Physical Models	6	
AM3530	Numerical Methods 2	6	
AM3540	Inverse Problems	6	
AM3550	Graph Theory	6	
AM3560	Advanced Probability	6	
AM3570	Fourier Analysis	6	
AM3580	Differential Geometry	6	
AM3590	Topology	6	

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Bachelor TW 2023

Contact for students	Pay attention! The answers to many of your questions are already on the web (https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/)! Are you a first-year student and did not you find the answer to your question on the website? Then, contact your mentor first. For higher-year students, the first person you may contact is the commissioner of AM (cow@ch.tudelft.nl). He or she knows answers to many questions, but can also refer to or ask other people from the degree program. During your entire studies, there are regular program meetings: In the first year there are quarterly meetings, in the higher-year semesters. During these meetings you will find all the important information about the program. They are announced on Brightspace and are also in the schedule. For questions about help and guidance in case of personal problems or studies, all students can contact study advisors. Also check the website here: https://www.tudelft.nl/studenten/faculteiten/ewi-studentenportal/organisatie/studieadviseurs/. For answers about specific subjects, first look at the Brightspace site of that subject and if you can not find the answer, contact the responsible teacher of that subject. Do you want to participate in improving the education? Then sign up for the course response group of your year. Contact your commissioner for education about this.
Program Coordinator	Bachelorcoordinator: Dr. J.A.M. de Groot J.A.M.deGroot@tudelft.nl
Term of validity	Pay attention! The information you find below is informative in nature and not binding! Binding information can be found in the rules and regulations (https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/regulations/). Keep an eye on Brightspace for changes and attend the program meetings for the latest information.
Program Goals	The graduate of the Bachelor's degree in Technical Mathematics 1. Is familiar with existing knowledge in the area of mathematics and its applications and has the competence to extend this by study. 2. Can, under supervision, develop own mathematical knowledge and insight in a purposeful and methodological manner. 3. Is able to choose mathematical models for analyzing problems from another discipline, and to validate their usefulness in practice. 4. Has developed a mathematical and analytical approach for solving problems and interpreting results, has a critical attitude and has insight into mathematical modeling. 5. Has knowledge and understanding of mathematical reasoning and deductive derivation along with formal mathematical proofs and can apply it generically. 6. Can work in a team and can communicate mathematical knowledge verbally and in writing to specialists and non-specialists. 7. Is aware of the connection between different scientific disciplines and their role in society.
Program Structure 1	The program AM lasts three years (3 x 60 EC). The first year consists of 9 study units each of 6 EC and 2 units of 3 EC. One of the subjects of 6 EC is an elective course. The second year consists of 9 study units of 6 EC and two study units of 3 EC. Also in the second year you have one elective course. The third year differs considerably. The first half of it follows a minor (see below). Then there are 2 elective courses of each 6 EC and the last 18 EC spent that year on your graduation project (also called a bachelor project). Each year is divided into 4 quarters of 10 weeks, where you follow courses worth of 15 EC. That means you have to spend 40 to 42 hours in your study every week. Of course, you may have to do more in one week than in another. It is therefore important to look ahead and plan your work. At AM we have quarterly courses and semester courses. A quarter course lasts 10 weeks and is usually 6 ECs. That means you have to spend about 17 hours a week. One semester takes two quarters, so 20 weeks. All semester courses are 6 EC, meaning that you spend about 8 hours a week on a semester course. Not all hours have been scheduled, you have to work a lot independently by yourself (self study). Sometimes you get homework to be done during the self study time, but you often have to think and decide yourself what you need to do to understand and remember the study material.
Program Structure 2	AM students and their teachers are part of the academic community of TU Delft. Each community has rules of conduct and TU Delft has laid down this in an ethical code (Code of Ethics). The following rules come from this code: Teachers and students: - have respect for others; - act involved, transparent and integer; - contribute to an inspiring work and study environment; - trust each other and avoid conflicts of interests; - respect each other's property and resources and those of the university Teachers: - behave righteously and respectfully with each other and students - aim for high quality and always strive for improvement; Students: - be fair and respectful towards each other and university staff; - get the best out of themselves by actively participating in their education and extra curricular activities; - encourage each other and their teachers by asking analytical questions and conducting well-argued discussions;
Exam requirements	The exam requirements state when you qualify for a AM diploma. You will find them in the Rules and Regulations of the Board of Examinors (https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/regulations/). In summary, you get a AM diploma if you have completed all the courses of the program, including the minor, with as a result a V (sufficient) or a grade of 6 or higher. You must apply for your diploma through the form on the forms website: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/forms/.
With Honours Regulation	Students who complete the program AM (3 years), have an minimum average of 8 for all courses, and have a minimum of 8 for

the bachelor project, receive their diploma with the mark 'with honours' (also: cum Laude).

The exact requirements can be found in the Rules and Guidelines of the Board of Examinors (<https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/regulations/>).

Minor Semester

In the first half of your third year you follow a minor. The minor is meant to give you the opportunity to deepen, broaden your knowledge and to obtain a foreign experience.

More information can be found at minors.tudelft.nl.

Gives access to

The bachelor AM gives access to TU Delft Master Applied Mathematics. You also have access to the masters of the same name at the University of Twente and Eindhoven University of Technology. To be admitted to the mathematics master's programs from other Dutch and foreign universities you may need to do a switching program. With a switching program, it is also possible to be admitted to non-Mathematics Master programs.

Expected prior Knowledge

The admission requirements for the AM program are: <https://www.tudelft.nl/en/education/programmes/bachelors/tw/bachelor-of-applied-mathematics/admission-requirements/>

The AM program advises students, who scored less than 8 for Maths B and 7 for English, NOT start the program. In recent years, it appears that these students have insufficient knowledge of English and Mathematics to complete the courses of the first year.

Education Methods

There are many different types of education at AM. The teachers try to convey the material in the most appropriate way. The most common forms of education are briefly described below:

- Class Lecture: At a class lecture a teacher gives a lecture in a classroom. Often this is supplemented with homework assignments.

- Instruction: An instruction is an exercise class. Under the supervision of a teacher and / or student assistants, you will work on assignments. Usually you are in an instruction with only a part of your fellow students in one room, so the group does not get too big and you get the chance to ask questions.

- Colstruction: A colstruction is a combination of a class lecture and instruction. Alternating, the teacher will explain and you work by yourself or groups.

- Practicum: During a practical session you (usually self-employed) do programming assignments and practical exercises under the guidance of student assistants.

- Project: A project is a practically oriented exercise where you make a big assignment in a group. Usually, something has to be delivered at the end of the project. Working together in the group is an important part of the project. At AM you will encounter this form of work at the Modeling courses in the first and second years.

- Self-study: this seems a stranger in the midst, but it is indeed the most important form of education! It is important that you get started yourself by all forms above. Only in this way make sure that you can apply everything that you have been taught by yourself.

Honours track

The Excellence Programme in the bachelor Applied Mathematics is offering a variety of extras to our best students. In each year, the extras represent 11-12 ECTS in addition to the regular courses. In the first two years, a number of courses offer an excellence programme module, presenting extra challenges to the students. In the second year, the practical assignments are more comprehensive and include on-site work with industry or other partners. In the third year, the Excellence Programme students take one specific extra Mathematics elective and 5 ECTS on interdisciplinary modules from the TUD Honours Programme.

You can apply for the Excellence Programme when your grade for the VWO subject Mathematics is at least 9 and you have an average of at least 8 for all subjects. If you meet both conditions you will be asked to submit a reference letter and a motivation letter, and you will be invited for an interview. More information can be found on <https://www.tudelft.nl/en/education/programmes/bachelors/tw/bachelor-of-applied-mathematics/excellence-programme-applied-mathematics/>

For good students looking for additional challenges, there is the Honors Program Bachelor (HPB) from the second year. It is accessible to all students who completed the first year in one year and had a minimum of 8 for all subjects on average. Students with an average of 7 can apply for admission.

More information can be found here: <https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/honours-programme/>

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Applied Mathematics

1e jaar BSc TW 2023

Program Coordinator	Bachelorcoordinator: Dr. J.A.M. de Groot J.A.M.deGroot@tudelft.nl
Prerequisites	There are no entry requirements for study components of the first year.
Introduction 1	Enrollment: For the courses AM1050-A Modeling-A and AM1050-B Modeling-B you have to enroll beforehand in Brightspace and choose a topic. For the (partial-)exams of all courses, you have to register via OSIRIS ultimately one weeks before the day of the exam. For more information, look at https://www.tudelft.nl/en/student/education/examinations/ . Do you not want to participate in a study component for which you have enrolled? Please unenroll yourself via Brightspace/OSIRIS!
Introduction 2	In the third quarter the first year elective course is scheduled. You can choose between TN1531TW Mechanics and Relativity, TI1520AM Algorithms and Data Structures, and EE1510AM Electricity and Magnetism. You can also choose for a different technical oriented course of 6 EC. Keep in mind that it is very likely that the schedule of this course will overlap with the schedule of your other courses. If you do consider to choose another course, then it has to be approved by the Board of Examiners. Therefore, contact the bachelor coordinator Joost de Groot on time (early in the second quarter).
Introduction 3	As with all bachelor programs at TU Delft, at Applied Mathematics there is a norm of 45 EC for a binding study advice (BSA). For more information, see http://bsa.tudelft.nl .
Introduction 4	Outlook year 2: The second year consists of two semesters. The first semester has 1 semester course and 4 quarter courses, 2 in each quarter. The second semester has 3 semester courses and 2 quarter courses, 1 in each quarter. The quarter course in the third quarter is an elective course. More information about elective courses can be found at the study guide web pages of the second year and will be presented at the quarter meetings.

AM1010	Mathematical Structures	6
Responsible Instructor	Dr. F.J. van de Bult	
Contact Hours / Week x/x/x/x	2/2/0/0 hc; 2/2/0/0 wc	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	Essentially none.	
Course Contents	A basic knowledge of high school mathematics is assumed and is necessary for some of the exercises. This course covers basic mathematical concepts, such as the structure of a proof, the natural numbers and induction, the real numbers and completeness, relations and functions, and the convergence of sequences and series.	
Study Goals	At the end of the course the student is able to * write and check the validity of simple mathematical proofs * apply and state the definitions of the basic concepts from the study of sets, relations, functions and cardinality; * prove statements using induction; * define and apply the concepts of supremum and infimum; * determine convergence of sequences and their limits using the definition or some well-known criteria; * determine convergence (absolute and relative) of a series using the definition or well-known criteria;	
Education Method	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) One lecture and exercise class each week. After the lecture there are introductory digital exercises in GraspLe. Subsequent homework problems are taken from the book. Each week you can optionally hand in (approximately) two exercises for formative feedback.	
Literature and Study Materials	Analysis with an Introduction to Proof: International Edition, 5/E Steven R. Lay ISBN: 978-1-78434-282-1 Pearson Paper, 400 pp. This is a special Delft University edition available at the study association CH. There are also some handouts available on the course website. Some important handouts have been included in the Syllabus for Analysis 1 as an extra service.	
Books	See Literature	
Assessment	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) The regular assessment consists of four parts - A 1 hour test in week 1.6 (T1) 10% or 0% - A 2 hour test in week 1.10 (M) 32% or 36% or 40% - A 1 hour test in week 2.5 (T2) 10% or 0% - A 2 hour test in week 2.10 (F) 48% or 54% or 60% The tests in weeks 1.6 and 2.5 count as a bonus, either both or just one, so the grade will be $\max(M*2/5 + F*3/5, \\ M*8/25 + F*12/25 + T1*1/10 + T2*1/10, \\ M*9/25 + F*27/50 + T1*1/10, \\ M*9/25 + F*27/50 + T2*1/10)$ It is not possible to retake one of these intermediate tests. The only resit possibility is a full resit for the entire course lasting 3 hours. At this occasion any previous results will be ignored. The results of the intermediate tests can also not be retained in subsequent years.	
Permitted Materials during Tests	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Pen and paper.	
Judgement	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) The regular assessment consists of four parts - A 1 hour test in week 1.6 (T1) 10% or 0% - A 2 hour test in week 1.10 (M) 32% or 36% or 40% - A 1 hour test in week 2.5 (T2) 10% or 0% - A 2 hour test in week 2.10 (F) 48% or 54% or 60% The tests in weeks 1.6 and 2.5 count as a bonus, either both or just one, so the grade will be $\max(M*2/5 + F*3/5, \\ M*8/25 + F*12/25 + T1*1/10 + T2*1/10, \\ M*9/25 + F*27/50 + T1*1/10, \\ M*9/25 + F*27/50 + T2*1/10)$ It is not possible to retake one of these intermediate tests. The only resit possibility is a full resit for the entire course lasting 3 hours. At this occasion any previous results will be ignored. The results of the intermediate tests can also not be retained in subsequent years.	
Co-Instructor	Dr. C. Kraaikamp	

AM1020	Kaleidoscope	6
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	2/0/0/0 hc; 2/0/0/0 wc; 2/0/0/0 pr;	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	The Kaleidoscope course consists of two parts, which are taught simultaneously. In the first part some basic subjects will be highlighted: - limits and continuity, - differentiation and integration, - complex numbers, - ordinary first order differential equations and - second order linear differential equations. In the second part you will get acquainted with the three software packages: - LaTeX, - Maple and - Matlab. You will learn how to use LaTeX to present texts with formulas in a clear and easy to read manner and Maple and Matlab for symbolic and numeric calculations.	
Study Goals	At the end of the course you are able to: - calculate limits, - apply different differentiation and integration techniques, - investigate whether a function is continuous/differentiable on its domain, - write complex numbers in different ways and use this to add/subtract/multiply/divide them and solve equations, - use different techniques to solve first order differential equations, - solve second order, linear differential equations. Further you have obtained basic skills in LaTeX, Maple and Matlab.	
Education Method	Instruction (4 hs/wk, computer lab (2 hs/wk) and self-study.	
Reader	https://openstax.org/details/books/calculus-volume-1,2 and 3	
Assessment	Further information can be found on Brightspace. There will be written exams for the theoretical part halfway and at the end of the course. Handing in computer (group) assignments is mandatory. The final grade will be calculated as follows: 100% is determined by the grades for the written exams. If both written exams have a grade greater or equal 4.0 then the final grade is the mean of both grades else the minimum of them. Further information can be found on Brightspace. The final mark for the course is the mark for the exam, but it is only valid when all computer assignments have been completed with sufficient results. In case of insufficient results for the computer assignments, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Judgement		
Co-Instructor	Dr. C. Kraaikamp	

AM1030	Linear Algebra 1	6
Responsible Instructor	Dr. J.A.M. de Groot	
Contact Hours / Week x/x/x/x	4/0/0/0 hc; 4/0/0/0 wc	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Required for	Many courses later on will use concepts and techniques of linear algebra 1, especially analysis 2, introduction to probability, algebra 1, ordinary differential equations, partial differential equations, numerical methods, systems theory, introduction to statistics, optimization, complexe function theory, lineare algebra 2, applied algebra, markov processes.	
Expected prior knowledge	Wiskunde B VWO	
Course Contents	Vectors are probably familiar: they are often presented as arrows in the plane or in space and are used to indicate a length and a direction. Vectors can be added and/or multiplied with a number. The course Linear Algebra studies vectors. First another representation of the arrows is discussed, making calculations with vectors much easier. One of the advantages of this representation is that it can be done more generally, which leads to the introduction of the vector space $\mathbb{R}^n$. A vector can then be imagined as a column (or row) consisting of an n-tuple of real numbers. Later (in Linear Algebra 2) it will be shown that it can be generalized even more: functions for example can also be viewed as vectors, and also vectors over for example complex numbers can be covered. The concept of vector space describes all of this. We are interested in vector spaces, or subspaces of them, that can be described using finitely many vectors. The reason is that calculations become much less tedious then. We will study when and how this is possible. In this context core concepts are (in)dependency, basis and dimension. The calculation tool used is solving systems of linear equations by row reduction. Using this method of solving systems, matrices are introduced. Later it will become clear that matrices are actually representations of so-called linear transformations between finite dimensional vector spaces. You can perform calculations with matrices: adding, scalar multiplication, multiplication, transposing and (if possible) inverting are operations that you should eventually master. In the study about the structure of matrices the concepts of eigenvalue and eigenvector are naturally revealed. These are for example used to find a very useful matrix decomposition that has many applications, for example in solving systems of differential equations or in understanding stochastic processes (processes where randomness plays a role). To also answer geometrical questions (for example finding points in subspaces that are as close to a given point as possible) the concept inner product is defined. This can be used to determine distances and angles between vectors. Core concepts are orthogonal basis, orthogonal complement, orthogonal projection and the least-squares method. Another concept (that has both a geometrical and algebraical interpretation) is the determinant of a square matrix. This is for example a measure for the volume enlargement with linear transformations, but it also indicates whether or not a matrix (or linear transformation) is invertible. Linear Algebra knows many applications, both in fundamental mathematics (algebra, probability theory) as in applied mathematics (numerical mathematics, differential equations, statistics, systems theory) as well as in physics (quantum mechanics). This is why it is one of the most important basic courses of the program.	
Study Goals	These can be found on the brightspace-pages of the course Linear Algebra 1 and in the dutch version of this site.	
Education Method	18 Lectures, 10 instructions. Most of the instructions are about two lectures. For a full description see the dutch version.	
Literature and Study Materials	Book: Linear Algebra written by Fraleigh and Beauregard (publisher Addison Wesley). Slides used during the lectures (they will be published on brightspace). Extra extra exercises (they will be published on brightspace). "AM1030 Lineaire Algebra 1 voor Technische Wiskunde Editie 2022". In this workbook the slides, extra exercises and two theoretical additions on the book are collected. The slides on brightspace contain a lot of extra information (examples, proofs), making the sheets a sort of "digital book". This extra information however is not easily printed, which is why the workbook only contains the slides without additional information. The left pages are empty, allowing you to put down your notes there. In the workbook the collections of extra exercises are also added. Answers, and sometimes also hints and explanations, will be regularly posted on brightspace.	
Assessment	2 partial exams (written): in week 5 two hours with weight 40%; in week 9 or 10 two hours with weight 60%. The final grade is the weighted average of the two unrounded grades. Condition: the unrounded grade of the 2nd partial exam must be at least 4.0. Otherwise the final grade is the minimum of the weighted average of the three unrounded grades and the grade of the 2nd partial exam. The resit (written) is short after the christmas break (or maybe in week 2.10), takes 3 hours and has a weight of 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. J.H. Weber	

AM1040	Analysis 1	6
Responsible Instructor	Prof.dr. J.M.A.M. van Neerven	
Contact Hours / Week x/x/x/x	0/4/0/0 hc; 0/4/0/0 wc; 0/2/0/0 mentoraat	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	This course builds upon the first part of the course Mathematical Structures (AM1010) and its contents are closely linked and interwoven. Before taking this course you should be familiar with the basic concepts of logic and proof, sets and functions, the real numbers and sequences (Chapters 1–4 in Lay).	
Course Contents	This is an introduction to Analysis for functions of one real variable. The following topics are treated: - limits - continuity - differentiation - integration - power series - the exponential and trigonometric functions	
Study Goals	After taking this course the student should be able to 1. understand and be able to reproduce the definitions and theorems related to the course contents, and prove the theorems 2. solve problems related to the course contents.	
Education Method	Lectures and exercise classes	
Reader	Lecture Notes Analysis 1	
Assessment	There are two written partial exams. The midterm exam takes 1 hour and counts 25% of the final grade while the final exam takes 3 hours and counts 75% of the final grade. If the student does not pass, they have to do a resit exam which fully determines the final grade (the partial exam will not contribute to the final grade in this case). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	No materials are permitted	
Judgement	There are two ways to obtain a final mark for the course Analysis 1, AM1040: 1. The final grade is the determined by the grades of the partial exams, the first counting 25% and the second counting 75% of the final grade. OR 2. The final grade is the grade of the resit. Earlier partial exam results do not count towards the final mark. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr.ir. R. Versendaal	

AM1050-A	Modelling A	3
Responsible Instructor	Prof.dr.ir. M.B. van Gijzen	
Contact Hours / Week x/x/x/x	0/0/4/0 prac	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	In this course the modelling cycle is presented. Students are given a simple but realistic problem. To solve this problem, a mathematical model should be made and a solution computed. After that the model should be evaluated to see if it describes the practical problem sufficiently well, or, if the model is not good enough, should be adapted.	
Study Goals	At the end of this course you will understand and be able to apply the modelling cycle.	
Education Method	Group work, personel coaching, programming in Matlab or Phytan	
Literature and Study Materials	All necessary documents are published on Brightspace	
Assessment	Written report and oral communication In case of an insufficient final result, a repair option exists in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The repair option is: submit an improved report Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Judgement	The course is evaluated per group. The performance during the lab sessions and the written report are the input for the grade. The quality of the research counts for 50 % of the grade. The process and the report each count for 25 % of the grade. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. M. Keijzer	

AM1050-B	Modelling B	3
Responsible Instructor	Dr.ir. M. Keijzer	
Contact Hours / Week x/x/x/x	0/0/0/4 prac	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Expected prior knowledge	The student has basic programming skills, e.g. as acquired in the course TW1050-A.	
Course Contents	In this course a start is made in working with the cycle of mathematical modeling. The student is presented with a not too complicated practical problem. After a mathematical model has been made, the problem is solved, sometimes using methods that are new for the student. The solution is evaluated with real data. If the comparison is not good enough, the model has to be adapted. In a second phase, the problem is expanded in a direction chosen by the students.	
Study Goals	Understanding and being able to apply the modeling cyclus. The use of tools such as a programming language and software. Communication with other students and with the tutor. Working in a team. Communicating about the research project and its results both verbally and in writing.	
Education Method	Working in groups of four students with personal coaching by a tutor.	
Literature and Study Materials	All necessary texts are posted on Brightspace.	
Assessment	Communications with the tutor A poster (or online pitch) on the research results and oral presentations at the poster/pitch session A research report A evaluation report with peer review In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Judgement	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) The grade is based on four partial grades, as determined by the tutors using rubrics: A grade for the modeling, based on the contents of the research report (40%). A grade for working as a team on the project, based on conversations with the tutor and the evaluation report. (30%) A grade for the poster (or online pitch), based on the form of the poster (or the slides) and the quality of the oral presentations at the poster/pitch session. (10%) A grade for the form of the research report. (20%)	
Co-Instructor	Dr.ir. J.T. van Essen	

AM1060	Algebra 1	6
Responsible Instructor	Prof.dr. D.C. Gijswijt	
Contact Hours / Week x/x/x/x	0/0/2/2 hc; 0/0/2/2 wc	
Education Period	3 4	
Start Education	3	
Exam Period	3 4 5	
Course Language	Dutch	
Required for	Algebra 2, Applied Algebra: Codes and Cryptosystems	
Expected prior knowledge	Required: Linear Algebra 1, Mathematical Structures	
Course Contents	This course gives an introduction to the theory of groups. The integers (modulo n), the symmetries of a geometrical figure and the permutations of a set of objects are all examples of groups and can therefore be studied in a common framework using groups. In this course we study groups in an abstract sense, and also consider applications to combinatorics, number theory and geometry. Topics include: group axioms and basic properties of groups, modular arithmetic, symmetry groups and permutations, group homomorphisms, quotients and products of groups, finite abelian groups.	
Study Goals	By the end of this course, you should be able to: 1. Verify if group theoretic definitions and notions apply in a given example. For instance, verify if something is a group, a (normal) subgroup, a homomorphism, a group action. 2. Perform simple computations in groups (e.g. Euclidean algorithm, modular arithmetic, working with permutations, calculations in quotient groups) and more complex computations (e.g. determine center, commutator subgroup, orbits, stabilizers). 3. Analyse relations between groups (e.g. determine if groups are isomorphic or determine the (number of) homomorphisms between them). 4. Use concepts, examples and theorems covered in the course to prove new mathematical statements.	
Education Method	Lectures and Exercise classes	
Literature and Study Materials	Digital lecture notes will be provided on Brightspace.	
Assessment	The final grade of the course consists of the following components: - Bonus1: 1 hour written test mid Q3 (10% or 0%) - Midterm: 2 hour written test end Q3 (32% or 40%) - Bonus2: 1 hour written test mid Q4 (10% or 0%) - Final: 3 hour written test end Q4 (48% or 60%) The final grade is the maximum of the following two outcomes: (1) $0.1 \cdot \text{Bonus1} + 0.1 \cdot \text{Bonus2} + 0.32 \cdot \text{Midterm} + 0.48 \cdot \text{Final}$ (2) $0.4 \cdot \text{Midterm} + 0.6 \cdot \text{Final}$ Repair possibilities: - Resit exam (written, 100%) It is not possible to retake individual tests. The only resit possibility is a full resit for the entire course. At this occasion any previous results will be ignored. The results of the intermediate tests can also not be retained in subsequent years. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Pen and paper only	
Co-Instructor	Dr. J.A.M. de Groot	

AM1070	Analysis 2	6
Responsible Instructor	Dr. K. Kirchner	
Contact Hours / Week x/x/x/x	0/0/2/2 hc; 0/0/2/2/ wc; 0/0/2/2 mentoraat	
Education Period	3 4	
Start Education	3	
Exam Period	3 4 5	
Course Language	English	
Expected prior knowledge	Analysis 1 (TW1040), Linear Algebra 1 (TW1030), Mathematical Structures (TW1010)	
Course Contents	From Analysis in one dimension to Analysis in higher dimensions. The topic of this course is differentiation and integration of functions of several variables and vector-valued functions.	
Study Goals	At the end of this course the student is familiar with all relevant aspects of differential calculus of functions of several variables and vector-valued functions, as well as the relevant aspects of integration of such functions. Specific learning objectives: 1. Limits and continuity Euclidian metric, open/closed sets, boundary points of sets, series in $\mathbb{R}^n$, limits, continuous functions. The student is able to describe, recognize and apply the above notions. 2. Differential calculus Partial derivatives, gradient, directional derivatives, differentiable functions, linearization, Taylor polynomials, extreme points of functions, chain rule, implicit function theorem. The student is able to determine whether a function is differentiable, can calculate the derivative and apply this in various situations. 3. Basic integration theory Multiple integrals, iterated integrals, Fubini's theorem, calculation of integrals with polar coordinates/cylindrical coordinates/spherical coordinates, change of variables for multiple integrals. The student is able to calculate multiple integrals with the help of Fubini's theorem and change of variable. 4. Integrals over curves and surfaces Curves and surfaces given by equations or parametrization, arc length of a rectifiable curve, arc length parametrization of a curve, line integral of a scalar function, line integral of a vector field along a curve, surface integral of a scalar function, surface integral of a vector field, conservative vector fields. The student is able to parametrize simple curves and surfaces, and to calculate various integrals over curves and surfaces with the help of a given parametrization. 5. Integral theorems Divergence and rotation of a vector field, Green's theorem, divergence theorem, Stokes theorem. The student is able to formulate and apply these theorems.	
Education Method	Lectures and exercise sessions.	
Literature and Study Materials	Lecture Notes Analysis 2	
Assessment	Two voluntary mid-term tests. Preliminary examination after quarter 3, final written examination.	
Permitted Materials during Tests	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Nothing except pen and paper	
Judgement	The final grade of the course is calculated as the maximum between A, B, C and D, where $A = 0.1 * \text{test1} + 0.32 * \text{preliminary exam} + 0.1 * \text{test2} + 0.48 * \text{final exam}$ $B = 0.1 * \text{test1} + 0.36 * \text{preliminary exam} + 0.54 * \text{final exam}$ $C = 0.36 * \text{preliminary exam} + 0.1 * \text{test2} + 0.54 * \text{final exam}$ $D = 0.4 * \text{preliminary exam} + 0.6 * \text{final exam}$ In order to pass, the grade of the final exam has to be bigger or equal to 5.0. When participating in the retake exam, the grade of the retake is the final grade of the course. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. M.V. Gnann	

AM1080	Introduction to Probability Theory	6
Responsible Instructor	R.C. Kraaij	
Contact Hours / Week x/x/x/x	0/0/0/4 hc; 0/0/0/4 instr	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Expected prior knowledge	Calculus in one and more variables, series and their convergence.	
Course Contents	Aim of the course is to develop insight in the mathematical structure of randomness. The basic notions of probability theory are introduced. The following subjects will be treated 1. Probability space, elementary rules, conditional probability, independence, Bayes theorem. 2. Discrete and continuous random variables. Most important discrete and continuous distributions and the contexts in which they are used. 3. Expectation, variance, probability and moment generating function. 4. Multidimensional random variables, independence, covariance, conditional probability density and conditional expectation. 5. Markov's and Chebychev's inequality, Jensen's inequality. 6. Convergence in probability and the law of large numbers. 7. Convergence in distribution and the central limit theorem.	
Study Goals	1. The student is able to apply the basic rules of probability such as the partition theorem, Bayes theorem and rules for independent events. 2. The student is able to perform basic computations with discrete and continuous random variables such as computation of expectation, variance, probability and moment generating functions. The student is able to compute the distribution of a transformed random variable. 3. The student is able to recognize which distribution to use in a concrete modeling context. The student can recognize distributions from probability and moment generating functions. 4. The student is able to compute expectations, marginal distributions, conditional distributions in the context of multivariate discrete and continuous random variables. The student is able to compute with covariance and apply the connection with independence. 5. The student is able to apply basic inequalities such as Markov, Chebychev and Jensen's inequality. The student can prove that a sequence of random variables converges by applying these inequalities. 6. The student is able to reproduce the proofs of the weak law of large numbers and the central limit theorem. 7. The student is able to prove the law of large numbers and central limit theorem under slightly modified conditions (such as weak dependence, not identically distributed).	
Education Method	Lecture and exercise classes	
Literature and Study Materials	Grimmett and Welsh, probability, an introduction, second edition, oxford university press, 2014 ISBN: 978-0-19-870997-8	
Assessment	Partial exam (counts for 0.4) and Exam (counts for 0.6). The resit is a single exam that counts for 1.0. For all three, one can take a single two-sided A4 with notes, written by hand. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	The book.	
Judgement	end-score= 0.4 (partial) + 0.6(exam) the result of the partial does not count for the resit disclaimer: information may change depending on the developments around the coronavirus.	
Co-Instructor	Dr. C. Kraaikamp	

AM1090	Introduction to Programming	6
Responsible Instructor	Drs. P.R. van Nieuwenhuizen	
Contact Hours / Week x/x/x/x	0/6/0/0 hc and 0/4/0/0 pr or 0/2/0/0 hc and 0/8/0/0 pr	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	There is no expected prior knowledge.	
Course Contents	In this course you will learn basic programming concepts and you will apply these concepts by writing your own computer programs in the programming language Python. During the lectures and the programming assignments the following subjects will be handled. - The interactive development environment IDLE. - Interactive use of IDLE and scripting. - Variables, assignment statements. - Input and output using keyboard and screen. - Data types integer, float and boolean. - Scopes, name spaces and modules. - Arithmetic and logical expressions. - Conditional statements. - Strings, lists, tuples, dictionaries and sets. - Indexing, mutable and immutable data types. - Loop constructions. - Definition of functions and function calls. - Recursive functions. - Use of modules and definition of modules created by yourself. - File input and output. - Data formatting for table output. - Use of numpy, matplotlib and the data types array and matrix. - Object oriented programming: classes, methods, constructors, inheritance and operator overloading. These subjects will be illustrated using examples from mathematics mainly. Also during the lab sessions many assignments are related to mathematics.	
Study Goals	After finishing this course you must be able to: - translate a description of a simple problem into an algorithm. - implement a simple algorithm in the programming language Python, using the following programming language constructions: logical and arithmetic expressions, assignment statements, functions, conditional statements, loop constructions, data structures with (nested) lists and other data types. - implement programs, which read input from files and write output to files. - define recursive solutions to problems and implement these solutions. - use Python with the libraries numpy and matplotlib for tasks like solving systems of linear equations and plotting graphs. - format program output in outlined columns in order to show the output on screen or to store it in files. - explain the benefits of object oriented programming. - implement your own classes, using inheritance and operator overloading.	
Education Method	Lectures (alternately 4 and 2 hours per week) and programming assignments (alternately 4 and 8 hours per week). Computer examination with two partial exams.	
Literature and Study Materials	To prepare for the (partial) exams, it is necessary to study the following material: - book: Think Python, Allen B. Downey, second edition (using Python 3) - all lecture slides - your own implementations of all lab assignments The book is online available in pdf and html format on the website www.thinkpython.com The lecture slides will become available in Brightspace shortly before or after every lecture.	
Assessment	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	A pdf version of both the book and the lecture slides is available on the computer during the test and exam. It is also allowed to use - A paper version of the book "Think Python". It may be a book bought from the bookshop or a copy of the complete book. No extra material may be added, like implementations of the lab assignments. - The printed lecture slides, also without extra material. Both the book and the lecture slides must be bound or stapled.	
Co-Instructor	Drs. I.A.M. Goddijn	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Applied Mathematics

keuzevakken 1e jaar TW 2023

AM1510EE	Electricity and Magnetism	6
Responsible Instructor	Dr.ir. B.J. Kooij	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In this course we make an introduction with the aid of Analysis and Vector-Analysis in Electricity and Magnetism. The subjects we will deal with are: Introduction in vector analysis, Elektrostatics, the electric field, Coulomb's law, charge distribution, field lines, flux, Gauss's divergence theorem, curl of the electric field, electric potential, Poisson's equation, Laplace's equation, boundary conditions, work and energy of static fields, conductors, surface charge (distributions), force on a conductor, capacitor, induced surface charge, force and energy, electric field of a dipole, electric field in matter, polarisation, induced dipoles. Magnetostatics, magnetic fields, magnetic force, electric current, Biot-Savart's law, magnetic fields due to a stationary current, divergence and curl of a magnetic flux, Stokes's theorem, Ampere's law, boundary conditions, Faraday's law, Maxwell equations.	
Study Goals	Elektriciteit en Magnetisme is een vak waarbij de student kennis maakt met de mathematische Fysica. De nadruk in deze cursus ligt meer op de wiskundige benadering van elektriciteit en magnetisme dan op een empirische natuurkundige benadering. Na de cursus is de student in staat om: 1) veldberekeningen te doen ten gevolge van lading en stroom verdelingen. 2) elektromagnetische configuraties te vertalen in wiskundige modellen.	
Education Method	Lectures and instruction	
Literature and Study Materials	Introduction to Electrodynamics, David J. Griffiths, Fourth Edition, Pearson Education; ISBN 10: 1-292-02142-X; ISBN 13: 978-1-292-02142-3	
Assessment	Handout (available on BB) "Electromagnetic Fields - local laws" by Dr. Martin D. Verweij Partial exams, week 5 and after week 10 or Final exam after week 10 The mean of both partial exams (weighting 50%) yields the grade of the course. The student has the choice after week 10 to do the final exam (3 hours) instead of the second partial exam (2 uur), where the result of the first partial exam is cancelled. Disclaimer: deze informatie kan veranderen op basis van onvoorziene omstandigheden of maatregelen (zie: OER Art. 29, lid 4)	
Permitted Materials during Tests	You can use a normal calculator during the exam.	
Judgement	The final grade is calculated by taking the mean of the not rounded grades of the partial exams, or by the grade of the final exam or the resit.	
Co-Instructor	Prof.dr. O. Yarovyj	

AM1520TI	Algorithms and Data Structures	6
Responsible Instructor	S. Hugtenburg	
Contact Hours / Week x/x/x/x	0/0/2/0 hc 0/0/4/0 instr.; 0/0/4/0 pract.	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Algorithms and data structures are fundamental notions in computer science. Understanding how they can be exploited in combination for better programming implementations in terms of time and space complexity is vital for writing efficient code. This course enables the student to: Understand, explain, and implement standard data structures. Understand, explain, and implement standard algorithms. Apply standard data structures and algorithms to solve programming tasks. Analyse and compare implementations with respect to their time and space complexity. Understand, explain, and apply standard programming techniques in the context of standard data structures.	
Study Goals	After this course, the student is able to: LO-1 define big-Oh, Omega and Theta notation for complexity of code. LO-2 analyse the run time complexity of (recursive) (pseudo)code. LO-3 analyse the space complexity of (recursive) (pseudo)code. LO-4 define hash and equality functions for custom classes in python. LO-5 analyse and implement (doubly) linked lists, arraylists, stacks, and queues. LO-6 describe and implement a variety of sorting algorithms (bubble sort, selection sort, merge sort, quicksort). LO-7 analyse and implement search trees, such as binary search trees, AVL trees, and red-black trees. LO-8 analyse and implement priorityqueues and heaps. LO-9 analyse and implement hash sets and maps. LO-10 prove claims about trees and graphs and algorithms operating on them. LO-11 analyse and implement graph algorithms, such as: Breadth-First Search (BFS), Depth-First Search (DFS), Dijkstra, and Prim. LO-12 select the right datastructure for a given problem with certain run time and memory requirements. LO-13 describe the difference between the problem classes P and NP and prove a problem to be part of P and/or NP.	
Education Method	This course features three different types of contact hours: - Lectures during which we discuss the material from the more theoretical point of view. - Instructions during which we discuss the implementation and more technical parts of the course. Bring a laptop as we will often do some live-coding during these sessions. - Labs during which there will be Teaching Assistants available to help you with any and all questions you have about the lab work. Additionally there are weekly sets of exercises. These exercises have strict deadlines before which they need to be handed in, in order to receive feedback on them. If you do not hand in anything, you will not get feedback.	
Books	The following book is excellent to review the material, but not strictly required for the course: Data Structures and Algorithms in Python by Goodrich et al.	
Assessment	The final grade of the course consists of several different components, each of which is discussed below. Analysis exam Week 5 (25%) This first exam will definitely feature open questions answered on paper, possibly contain multiple-choice questions, and focus mainly on goals LO-1 to LO-6. The exam is to take 1.5 hours. The score for this exam is denoted as Ma. Implementation exam Week 5(25%) This exam features several implementation questions that are completed on a computer. The exam focuses mainly on goals LO-4 to LO-6 The exam is to take 1.5 hours. The score for this exam is denoted as Mi. Final implementation exam Week 10 (25%) This exam features several implementation questions that are done on a computer. The exam is about all learning objectives, with a larger emphasis on the material not covered by the midterm exam. The exam is to take 1.5 hours. The score for this exam is denoted as Fi. Final analysis exam Week 10 (25%) This exam will definitely feature open questions answered on paper and possibly contain multiple-choice questions. The exam is about all learning objectives, with a larger emphasis on the material not covered by the midterm exam. The exam is to take 1.5 hours. The score for this exam is denoted as Fa. Final grade The final grade for the course is computed as follows: $G = 0.25Ma + 0.25Mi + 0.25Fi + 0.25Fa$. If $(0.5Ma + 0.5Fa) < 5$, the final grade is insufficient. If $(0.5Mi + 0.5Fi) < 5$, the final grade is insufficient. Else the final grade is G. There are two resits of 3 hours in the exam week. Ra and Ri and they replace $(0.25Ma + 0.25Fa)$ and $(0.25Mi + 0.25Fi)$ with $0.5Ra$ and $0.5Ri$ respectively, if they increase the final grade. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	I.D. van Kreveld	

AM1531TN	Mechanics and Theory of Relativity	6
Responsible Instructor	Dr. N. Chepiga	
Contact Hours / Week x/x/x/x	4 hours lectures and 4 hours work lectures	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Calculus 1	
Course Contents	Classical Mechanics and Special Relativity Topics: Newton's laws, force laws, conservation of energy, momentum and angular momentum, collisions, rotating bodies. Special Theory of Relativity Topics: Einstein postulates, time dilation and length contraction, Galilean and Lorentz-transformations, four-vectors, energy-momentum vector and $E=mc^2$, forces and collisions at high velocities.	
Study Goals	at the end of the course, you should be able to: 1) Apply the basic laws of classical mechanics (CM) and special relativity (SRT). 2) Derive equations of motion for arbitrary situations by analyzing the forces present. 3) Apply the conservation laws of energy, momentum and angular momentum. 4) Formulate and prove Kepler's laws for planetary motion. 5) Relate inertial frames through Galilei (CM) and Lorentz (SRT) transformations. 6) Apply the concepts of time-dilation and length-contraction in SRT. 7) Describe physical phenomena at high speed through four-vectors. 8) Apply conservation of the energy-momentum four-vector to collisions in special relativity.	
Education Method	Lecture (2x2 hours weekly), tutorials (2x2 hours weekly), and self-study.	
Literature and Study Materials	T. Idema, Mechanics and Relativity, TU Delft Open Textbooks, ISBN 978-94-6366-085-3, available at https://textbooks.open.tudelft.nl/index.php/textbooks/catalog/book/14	
Books	T. Idema, Mechanics and Relativity, TU Delft Open Textbooks, ISBN 978-94-6366-085-3, available at https://textbooks.open.tudelft.nl/index.php/textbooks/catalog/book/14	
Assessment	Midterm (30%) and final exam (70%). Activity during lectures: up to 5% bonus The resit consists of a single written exam that counts for 100%.	
Co-Instructor	Dr. B. van den Dries	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Applied Mathematics

2e jaar BSc TW 2023

Program Coordinator	Bachelorcoordinator: Dr. J.A.M. de Groot J.A.M.deGroot@tudelft.nl
Prerequisites	Be aware! The entry requirements have been recorded in the OER. If there is a difference between what is listed here and what had been recorded in the OER, the content of the OER is leading. To participate in the courses AM2050-A Modeling-2A (simulation) and AM2050-B Modeling-2B (differential equations), it is mandatory to have finished the courses AM1050-A Modeling-A and AM1050-B Modeling-B.
Introduction 1	Enrollment: For the courses AM2050-A Modeling-A and AM2050-B Modeling-B you have to enroll in Brightspace beforehand and choose a topic. For the (partial-)exams of all courses, you have to register via OSIRIS ultimately two weeks before the day of the exam. For more information, look at https://www.tudelft.nl/en/student/education/examinations/ . If you do not want to participate in a study component for which you have enrolled: Please unenroll yourself via Brightspace/OSIRIS!
Introduction 2	Special components year 2: In the third quarter you will have an elective course. You can choose from all elective courses that are listed in the study guide (courses from Leiden included), but the schedule of only the courses with a AM2-code have been aligned with the schedule of the other second year courses. To obtain your TW-diploma you need at least one elective course with a AM3-code, or an elective course from Leiden.
Introduction 3	Outlook year 3: In the third year you will do a minor in the first semester. You will choose this minor already in your second year. For more information, see http://minors.tudelft.nl . Because the content of minors may overlap with the content of the bachelor program Applied Mathematics, you need to have the minor of your choice approved beforehand by the bachelor coordinator Joost de Groot. Do you want to go abroad in your third year? Then start to inform yourself at https://www.tudelft.nl/studenten/ewi-studentenportal/organisatie/international-internship-office .

AM2010	Linear Algebra 2	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	4/0/0/0 hc: 4/0/0/0 wc	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Linear Algebra 1 (AM1030)	
Course Contents	The course consists of two parts: The first part of the course focuses on the concept of vector spaces and their applications. Topics to be discussed are: basis transformations, quotient spaces, linear operators and their properties, subspaces, eigenvalues and eigenvectors, the Jordan canonical form, inner product and normed vector spaces and their operators, projections, and quadratic forms. The second part of the course addresses concepts from numerical linear algebra. Topics to be discussed are: special matrix types, vector and matrix norms, orthogonal decompositions and orthogonalization via the Gram-Schmidt process, Gershgorin's circle theorem for estimating eigenvalues, as well as the singular value decomposition and pseudo inverses of a matrix.	
Study Goals	At the end of the course, you should be able to: + Construct finite dimensional vector spaces, subspaces, and inner product spaces as well as linear operators on those spaces. + Analyze finite dimensional vector spaces, subspaces, inner product spaces, and linear operators with respect to their properties via derivations and proofs. + Compute eigenvalues and (generalized) eigenvectors of linear operators respectively perform their (partial) diagonalization. + Compute the singular value decomposition of matrices and linear operators.	
Education Method	The course consists of lectures and practical sessions. The theoretical concepts will be explained during the lectures and illustrated by examples. The practical sessions are meant for reviewing the theoretical concepts and for applying them to examples, which is a good preparation for the exam.	
Computer Use	Basic experience with MATLAB (or Python) is recommended for checking the correctness of some computations of, e.g., eigenvalues/vectors, Jordan canonical forms, and singular value decompositions.	
Literature and Study Materials	Lecture slides will be provided	
Books	L. Sadun, Applied Linear Algebra. The Decoupling Principle, Second Edition, AMS, 2008.	
Assessment	A two-hours intermediate written exam in week 1.5 counting for 30% and a three-hours written end exam in week 1.9 counting for 70% of the final grade. If the grade of the end exam is <4.5 then the end exam counts for 100% of the final grade. A three-hours resit exam is offered in week 2.5 of the third quarter counting for 100% of the final grade.	
Permitted Materials during Tests	Non-programmable pocket calculator	
Co-Instructor	Dr. M. Möller	

AM2020	Optimization	6
Responsible Instructor	Dr.ir. L.J.J. van Iersel	
Contact Hours / Week x/x/x/x	0/4/0/0 hc 0/4/0/0 wc	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear Algebra 1 (AM1030), Analysis 2 (AM1070), Mathematical Structures (AM1010)	
Course Contents	Introduction to optimization, convexity. Modelling (Mixed) Integer Linear Programming problems. Linear programming: Simplex method, duality. Network optimization: shortest paths, max flow. Integer linear programming: branch & bound, Gomory cutting planes. Nonlinear optimization: Newton method, steepest descent. Complexity: NP-hardness. Approximation algorithms.	
Study Goals	The students are able to formulate discrete optimization problems as (mixed integer) linear programming problems. The students are able to execute basic algorithms for solving linear programming, integer linear programming and nonlinear programming problems, to understand the outcomes of the algorithms and to apply duality. The students know basic theorems of linear and nonlinear optimization and are able to prove variations of these theorems. The students can apply basic algorithms for network optimization problems (shortest path, max flow, travelling salesman problem). The students are able to design reductions between optimization problems and to prove correctness of those reductions.	
Education Method	4 hours lecture and 4 hours of exercise class per week, plus feedback exercises and homework exercises	
Literature and Study Materials	Lecture notes	
Assessment	Written exam.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	3 hours	
Permitted Materials during Tests	A non-graphical calculator	
Judgement	The exam grade is the final grade	
Co-Instructor	M.J. Jones	

AM2030	Ordinary Differential Equations	6
Responsible Instructor	Dr. Y. van Gennip	
Contact Hours / Week x/x/x/x	0/4/0/0 hc; 0/4/0/0 wc	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Analysis 1 (AM1040), Linear Algebra 1 (AM1030)	
Course Contents	Other topics from first year courses are needed as well, such as: -series from Mathematical Structures (AM1010); -complex numbers from Kaleidoscope (AM1020); -partial derivatives from Analysis 2 (AM1070). This course is about ordinary differential equations. For first order equations we will treat linear, separable, and exact equations. We will prove existence and uniqueness of solutions, using Picard iterations. For second order equations we will treat linear equations, power series solutions, and the method of Frobenius. The Laplace transform will also be treated, as will systems of linear differential equations. Stability and phase plane analysis will play an important role.	
Study Goals	After finalizing the course, you should be able to use the theory of ordinary differential equations (ODEs) to study (systems of) differential equations. In particular you should be able to: 1. Solve linear (systems of) ODEs. 2. Solve separable ODEs and exact ODEs. 3. Determine the stability of solutions of ODEs. 4. Prove or disprove local existence and uniqueness of solutions. 5. Solve second order ODEs using power series and Frobenius theory. 6. Solve ODEs using Laplace transforms. 7. Determine features (like periodic behaviour, asymptotic behaviour, boundedness) of (solutions of) nonlinear (systems of) ODEs and sketch their phase planes.	
Education Method	Lectures (4 hours a week) and exercise classes (4 hours a week)	
Literature and Study Materials	The book that will be used in this course is Differential Equations and their Applications, Fourth Edition. Martin Braun, Springer, 1993, ISBN: 0-387-97894-1	
Assessment	Additional materials may be provided during the course. There will be a single written final exam (100%). The resit consists of a single written exam (100%).	
Permitted Materials during Tests	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) You may use a graphical calculator at the exam. If necessary tables of Laplace transforms will be distributed.	
Co-Instructor	Dr. J.L.A. Dubbeldam	

AM2040	Complex Function Theory	6
Responsible Instructor	Dr.ir. W.G.M. Groenevelt	
Contact Hours / Week x/x/x/x	0/0/0/4 hc: 0/0/0/4 wc	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Expected prior knowledge	AM1010 Mathematical Structures, AM1020 Kaleidoscope (Complex Numbers), AM1040 Analysis 1, AM1070 Analysis 2	
Course Contents	This course offers an introduction to the theory of analytic functions of one variable. An analytic function is a complex-valued function defined on an open set in the complex plane that is complex differentiable on its domain. Unlike functions in a real variable, the mere existence of a complex derivative has strong implications for the properties of the function. At the center of the course is the Cauchy integral formula for analytic functions. This formula expresses the value of an analytic function inside a circle in terms of the values of the function on the circle. The integral formula allows us to derive numerous beautiful properties of analytic functions. The general goals of the course are: - Being able to extend the basic notions from real-valued function of a real variable to complex-valued functions of a complex variable and having an understanding of the similarities and differences between real and complex analysis. - Acquiring analytic skills that enable one to choose an applicable methods from Complex Analysis for problems from various fields such as mathematical physics, system theory, signal analysis and the skills to turn this choice into a solution of the given problem. The following topics will be covered: 1. The basic complex function and their properties: the exponential function, trigonometric functions and logarithms 2. Analytic functions, the Cauchy-Riemann equations, harmonic functions 3. The Cauchy integral formula with applications, such as Liouville's theorem and the maximum-modulus principle. 4. Power and Laurent Series 5. Classification of isolated singularities 6. The residue theorem with applications, such as the argument principle and Rouché's theorem.	
Study Goals	By the end of the course you should be able to 1. perform basic mathematical operations (arithmetics, powers, roots) with complex numbers in Cartesian and polar forms, and solve elementary complex equations; 2. describe mapping behaviour of complex functions, including multi-valuedness and branching; 3. find the Taylor and Laurent series of a function and determine its domain of convergence; 4. determine the isolated singularities of a function, as well as the type of the singularities; 5. evaluate contour integrals using parametrization, primitive functions, Cauchy's integral formula and the Residue theorem; 6. state and apply fundamental properties of analytic functions, including Liouville's theorem, the maximum-modulus principle, the Identity theorem, Rouché's theorem; 7. apply various techniques to evaluate (improper) integrals, including contour integration, ML-estimates, Jordan's lemma, shrinking path lemma; 8. explain concepts, state and prove theorems and properties involving the above topics.	
Education Method	Lectures and exercise classes	
Literature and Study Materials	Nakhlé H. Asmar, Loukas Grafakos - Complex Analysis with applications, Undergraduate Texts in Mathematics. Springer, 2018, ISBN 978-3-319-94062-5. This book is accessible online for TU-students via link.springer.com	
Assessment	Written exam.	
Judgement	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr. J.A.M. de Groot	

AM2050-A	Modelling 2A	3
Responsible Instructor	G.F. Nane	
Contact Hours / Week x/x/x/x	0/0/4/0 prac	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	a. Required: Introduction to statistics b. Strongly advised: Introduction to probability	
Course Contents	This modelling course focusses on applications of probability and statistics. Each group will work on an assignment handed out at the beginning of the course. Supervision consists of weekly meetings with the group's supervisor.	
Study Goals	Disclaimer: the study guide information may change depending on the developments around the coronavirus. By the end of the course, you should be able a. Apply standard statistical techniques to solve real-world like data-driven problems. b. Apply more advanced statistical techniques and compare those with the standard employed techniques. c. Analyse the results obtained from the standard and advanced statistical techniques and evaluate the performance of the methods. d. Introduce and motivate the choice of the statistical methods, and present and discuss results of the (performance) analysis in a report. e. Provide recommendations for the stakeholders course of action. f. Breakdown the problem in multiple tasks and distribute the work within the team. g. Present your results in an engaging and interactive way in a video.	
Education Method	-After an introductory meeting in the first week, there will be weekly meetings of each group with its supervisor. - Groups will produce midterm reports and receive feedback from supervisors. - At the end of the course, groups hand in the final report and summarize their findings with a short video.	
Literature and Study Materials	See Brightspace	
Assessment	Report (80%) Video (20%) In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Judgement	Based on the written report (80% of the final grade) and video with findings (20% of the final grade)	
Co-Instructor	Dr. D. Kurowicka	

AM2050-B	Modelling 2B	3
Responsible Instructor	Dr. H.M. Schuttelaars	
Instructor	Dr.ir. E.G. Rens	
Instructor	Dr. A. Geyer	
Instructor	Dr.ir. Y.M. Dijkstra	
Contact Hours / Week x/x/x/x	0/0/0/4 prac	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	n.a.	
Course Contents	The course focusses on modelling a practical mathematical physics problem.	
Study Goals	At the end of the course, the student 1. is able to develop a simple mathematical model for a given practical problem. 2. is able to apply these methods and techniques to a practical modelling problem. 3. is able to acquire additional knowledge that is relevant for solving a practical problem. 4. is able to apply the mathematical modelling cycle and comment on the usefulness and weaknesses of mathematical models. 5. is able to determine the consistency and the validity of a mathematical model by analysis of simple test cases. 6. is able to communicate with the problem owner (supervisor) to get relevant information necessary for solving the practical problem and can deal with the uncertainties in this information. 7. is able to determine the different sub problems and is able to solve the complete problem as a team. 8. is able to present the results of the project through an oral presentation and by writing a report. 9. is able to use tools as programming languages, mathematical software and application software. 10. has a better overview of the applicability of mathematics.	
Education Method	Working on the project in a group of studenten with weekly contact with the supervisor of the project	
Literature and Study Materials	n.a.	
Assessment	The assessment is based on the assessment rubric. Here we have the following aspects: - Research : Quality of work done in the project (30%) - Process: The way the work has been carried out (30%) - The report (30%) - The presentation (10%)	
Special Information	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) The study guide information may change depending on the developments around the coronavirus	
Judgement	Four grades for the aspects: - Research (30%) - Process (30%) - Report (30%) - Presentation (10%)	
Co-Instructor	Final grade is a weighted average of these four grades Dr. A. Geyer	

AM2060	Numerical Methods 1	6
Responsible Instructor	mr.ir.drs. V.N.S.R. Dwarka	
Contact Hours / Week x/x/x/x	0/0/2/2 hc; 0/0/2/4 wc	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Required for	Further numerical courses, modelling and BSc-project	
Expected prior knowledge	Analysis I and II, Linear Algebra I and II	
Summary	Introduction to numerical mathematics, rounding/truncation errors, interpolation, numerical differentiation, numerical time integration for initial value problems, boundary value problems (finite differences), heat equation, numerical integration (quadrature), non-linear equations.	
Course Contents	Numerical mathematics is the study of methods that give numerical approximations to mathematical problems. In most cases, the exact solution cannot be obtained (easily). During this course, numerical methods are treated for the solution of ordinary differential equations, and a relatively simple partial differential equation (the time-dependent heat equation). Further several numerical methods are treated which can also be applied to problems that do not involve differential equations. Such methods are: interpolation, numerical integration, and solution strategies for non-linear equations. The use of numerical software (MATLAB/Python) is an essential part of the course material.	
Study Goals	After having completed the course, the student will be able to design, analyse and implement numerical methods for ordinary differential equations. At the end of the course, the student will be able to: -design numerical methods for ordinary differential equations -analyse properties of the numerical methods (for example stability and consistency) -understand the numerical treatment of time-dependent and stationary ordinary differential equations and/or a combination of both -distinguish between the numerical treatment of a linear and non-linear problem -implement the numerical methods (for example using python or matlab)	
Education Method	Two hours of lectures and an afternoon of lab work/instruction per week: -During the instruction hour, you will be able to work on homework sets. These homework sets are for practice purposes and are not-graded. -During the lab instruction, you will be able to work on the lab assignment (programming).	
Literature and Study Materials	Numerical Methods for Ordinary Differential Equations, C. Vuik, F.J. Vermolen, M.B. van Gijzen, en M.J. Vuik (DAP / VSSD, 2015)	
Assessment	Lab assignment (pass/fail which grants access to exam) and a written test. Only a pass from the previous study year will be valid. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Pen, non-programmable calculator and workpaper (which will be provided on the spot)	
Judgement	The final grade is determined as follows: 100% written test. The final grade will only be awarded provided the lab report is satisfactory.	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

AM2070	Partial Differential Equations	6
Responsible Instructor	Dr. H.M. Schuttelaars	
Contact Hours / Week x/x/x/x	0/0/4/4 hc	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	The course Ordinary Differential Equations AM2030	
Course Contents	Quasi-linear first order PDEs, and a simple model describing traffic flow. Types of second order PDEs : parabolic, elliptic, and hyperbolic PDEs. Derivation of the wave equation. Initial and initial-boundary value problems. Waves and reflections. Fourier series and the method of separation of variables. Sturm-Liouville problems. derivation of the heat equation. Boundary value problems. Delta functions and distributions. Green's functions for heat, wave, and Laplace equations. Fourier and Laplace transform methods.	
Study Goals	At the end of the course, the student should be able to: 1. solve initial value problems and initial-boundary value problems for wave, transport, and heat equations. 2. solve boundary value problems for Laplace and Helmholtz equations. 3. make the right choice for the method to solve the problems listed in 1 and 2. 4. give a physical interpretation of the constructed solutions. 5. visualize the results by using the formula manipulation package MAXIMA.	
Education Method	lecture, exercise class, and practical work	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972	
Assessment	Home-work exercises, practical work, and a written exam The content of the courses WI3150TU and WI3151TU is the same as the course AM2070. However, the assessment of the course AM2070 is not the same as the assessment of the courses WI3150TU and WI3151TU. Therefore, by default, these courses are not interchangeable. Students may submit a motivated request to the Board of Examiners and request to be considered for an exception to this rule. "The final mark for the course is the mark for the exam, but it is only valid when all practical work have been completed with a pass. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	pen, pencil, passer, geo-triangle (tables will be handed out during the exam)	
Special Information	the study guide information may change depending on the developments around the coronavirus.	
Judgement	Final grading. 1) The practical work is mandatory, and as long as the practical work is not completed no final grading will be given. 2) Twice per year a written exam will be held. For the written exam in the fourth teaching quarter the following rules hold. The final grading is equal to the maximum of { the written exam grading, the average of the written exam grading and the home-work grading (where the exam grading has to be at least 5.0)}. For the written exam in the other period the final grading is equal to the written exam grading.	
Co-Instructor	Ir. W. van Horssen	

AM2080	Introduction to Statistics	6
Responsible Instructor	Dr. H.P. Lopuhaä	
Instructor	Dr. A.F.F. Derumigny	
Instructor	Ir. J.G. Rou	
Contact Hours / Week x/x/x/x	4/0/0/0 hc; 4/0/0/0 wc	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Introduction to Probability (first year probability course)	
Course Contents	Statistical Models; Descriptive Statistics, Estimation theory; Testing theory; Confidence regions; Regression	
Study Goals	By the end of this course: 1. You know the concept of statistical model and you are able to formulate such a model for standard datasets. 2. You know several numerical and graphical summaries of a dataset and their relation with the probability distribution in the statistical model for the corresponding dataset and you are able to produce them with the statistical package R. 3. You should know the concepts of Mean Squared Error and unbiasedness for an estimator and you are able to determine them for simple estimators. 4. You know the Maximum Likelihood, Method of Moments, and Bayesian approach towards estimation and you are able to construct estimators according to these principles and compare their performance either by hand or by means of a simulation with the package R. 5. You know the basic concepts involved in hypothesis testing and you are able to setup and conduct a statistical test in simple statistical models. 6. You know the setup of the Student-t test for one sample, for two paired samples, and two independent samples, and you are able to perform these tests either by hand or by means of the package R. 7. You know the setup of likelihood ratio tests and you are able to perform these tests either by hand or by means of the package R. 8. You know the concept of confidence interval and you are able to construct these intervals from pivots or near-pivots for simple statistical models. 9. You know the formulation of the linear regression model and the maximum likelihood estimators for the unknown parameters and you can fit these models by means of the statistical package R. 10. You know several concepts to check the assumptions and assess the fit of a linear regression model and you are able to determine these by means of the package R. 11. You know the content of important results, such as Theorem 4.29 and its proof, Theorem 4.43 and Theorem 5.9. 12. You know the definition of score function and Fisher information and their properties formulated in Lemma 5.10.	
Education Method	Lectures, exercise classes, and computer assignments.	
Computer Use	Statistical analyses are performed by means of the (free-ware) package R. See www.r-project.org	
Literature and Study Materials	"An Introduction to Mathematical Statistics", 1st edition. Fetsje Bijma, Marianne Jonker and Aad van der Vaart. Slides of the lectures are made available through Brightspace. Weekly written and/or computer assignments	
Assessment	Statistical Software package R and the program R-studio Mandatory R-assignments that have to be completed with a PASS. Written Midterm exam in week 5, final written exam in week 10 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	2 hour Midterm exam, 3 hour Final exam	
Permitted Materials during Tests	When the exam is held on campus, then the exam is a "closed book" exam. You are only allowed to use sheets with information on probability distributions and R-commands. You are not allowed to use any books or notes. When the exam is held online, then book and personal notes are permitted. NOT permitted is contact with third parties or getting information from the internet during the exam.	
Judgement	M: result of the midterm E: result final exam Final Grade after final exam: $0.6 \cdot E + 0.4 \cdot M$ Final Grade after resit: grade of the resit The course is completed only with a sufficient final grade for the written exam and if all R-assignments have been completed with a PASS	
Co-Instructor	Prof.dr.ir. G. Jongbloed	

AM2090	Real Analysis	6
Responsible Instructor	Dr. Ir. E. Lorist	
Contact Hours / Week x/x/x/x	4/4/0/0 colstr	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Required for	All subsequent courses in analysis and probability theory.	
Expected prior knowledge	Mathematical Structures, Analysis 1 and 2, Linear Algebra 1.	
Course Contents	The course consists of two parts. The first part of the course is about metric spaces. Metric spaces are a generalization of, e.g., the Euclidean space and are needed in applications to various other mathematical fields. Common examples are: - The space of continuous functions (Differential equations) - The space of square integrable functions (Partial differential equations, Numerical analysis, Probability theory, Quantum mechanics, Graph theory, etc.) - The space of summable sequences (Machine learning) You will learn about metric spaces during the course. Among other things, we will discuss: convergent sequences, open and closed sets, continuity and homeomorphisms, total boundedness and completeness, compactness, uniform convergence and spaces of continuous functions. The second part of the course is about measure and integration theory. This theory is at the foundation of a large part of modern mathematics. It is used in Fourier analysis, (partial) differential equations, probability theory and numerical analysis. In measure theory, we study how to measure a set in a fair way. We will be able calculate volumes and areas of sets that cannot be treated with the Riemann integration theory from Analysis 1 and 2. This will lead us to a larger class of functions which we can integrate and very strong convergence theorems. We will end the second part of the course with an application to Fourier series.	
Study Goals	At the end of the course, you should be able to write rigorous mathematical proofs in the context of metric spaces and Lebesgue integration theory. The first quarter focusses on metric spaces and the second quarter on Lebesgue integration theory. By the end of the first quarter, you should be able to LO1.1: Formulate the definition of the main notions of the theory of metric spaces and explain the connection to their counterparts in the real number line. LO1.2: Prove the characterizations of convergent sequences, open and closed sets and continuous functions treated during the course. LO1.3: Use the definitions of totally bounded, complete and compact metric spaces to prove properties of such spaces. LO1.4: Apply the definition of pointwise and uniform convergence to show results for sequences of continuous functions. By the end of the second quarter, you should be able to LO2.1: Apply the definition of a measure (in particular, the Lebesgue measure) and the notions it builds upon to prove basic properties. LO2.2: Use the construction of the Lebesgue integral for measurable functions and its connection to the Riemann integral for continuous functions to prove results for integrals. LO2.3: Prove and apply the three main convergence theorems (the monotone convergence theorem, Fatou's lemma and the dominated convergence theorem). LO2.4: Perform manipulations in L^p spaces, like applying Minkowski's and Holder's inequality and using completeness.	
Education Method	Lectures + Exercise classes	
Literature and Study Materials	Real analysis, N. L. Carothers, Cambridge University Press, Cambridge, 2000, ISBN 0-521-49756-5	
Assessment	Two 2-hour exams, one after each quarter. If both attained grades are at least a 5.0, the final grade is determined as the average of these grades. Four assignments, which count for 15% of the final grade if this increases the final grade. A 3-hour resit about both parts of the course, which directly determines the final grade. The assignments do not count for the resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Pen and paper	
Co-Instructor	Dr.ir. M.C. Veraar	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Applied Mathematics

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Program Coordinator	Bachelorcoordinator: Dr. J.A.M. de Groot J.A.M.deGroot@tudelft.nl

AM2510	Decision Theory	6
Responsible Instructor	G.F. Nane	
Contact Hours / Week x/x/x/x	0/0/4/0 colstr; 0/0/4/0 prac	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Probability theory and Statistics	
Course Contents	The course covers different aspects of the following subjects: Forecasting Inventory Management Judgmental Forecasting Rational decision theory Decision trees Influence diagrams Bayesian Networks	
Study Goals	1. Learn how to model and solve real-life decision problems. 2. Understand and master the theoretical concepts covered during the lectures or used during the projects. 3. Understand the methods taught during lectures in detail, as well as the assumptions and motivation. 4. Apply statistical methods to real life situations 5. Provide support in decision-making problems. 6. Are able to master basic software statistical tools and to find appropriate functions to use during projects. 7. Are able to communicate the results to the problem owner and can justify/motivate the undertaken steps.	
Education Method	Lectures & Projects Expected workload 1. Attending lectures and practicum: 72 hours 2. Working for the project assignments (apart from the practicum time): 80 hours 3. Preparing for the oral exam: 10 hours 4. Oral exam: 1 hour 5. Preparing lectures, homeworks, correcting other assignments: 5 hours	
Computer Use	R, Python, Matlab, Excel	
Literature and Study Materials	Lecture slides and notes (diktaat), online videos and materials (from MOOC)	
Practical Guide	4 project assignments	
Books	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) additional (not required) reading: "Decision Systems for Inventory Managment and Production Planning" E.A.Silver "Experts in Uncertainty" R.M Cooke	
Reader	Basic Concepts of Applied Decision Theory (diktaat)	
Assessment	Reports of projects & Oral exam	
Judgement	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Reports for 4 projects are judged from the perspective of : - correctness of analysis - clarity of presentation - innovativeness of proposed method Oral exam: Discussion of theory used in the projects, results obtained and conclusions drawn. - each report is graded and feedback is provided. For the first two reports, the students can improve their grade by 0.5 points, if they address all the comments provided in the feedback. - during the oral exam, the grade of each project is updated based on the answers of students during the oral exam.	
Co-Instructor	Dr. D. Kurowicka	

AM2520-H	History of Mathematics	6
Responsible Instructor	Dr. K.P. Hart	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	No specific prior knowledge, other than the content of the first three semesters year of the curriculum Applied Mathematics, is required. We do explicitly expect a wide interest in mathematics, its history, and its philosophy.	
Summary	Disclaimer: the study guide information may change depending on the developments around the coronavirus. Only on September 1st will it be in its final form.	
Course Contents	This course gives an overview of parts of the history of mathematics. Which parts may to some extent be dictated by the interest of the students, although certain 'standard' developments will be treated regardless. Among the 'fixed' topics are - developments in Egyptian and Mesopotamian mathematics - Greek mathematics - mathematics in the Indian and Arab worlds - how certain notions developed through time: calculus, algebra, geometry - the foundations of mathematics, then and now	
Study Goals	At the end of the course: 1. the student knows some of the major developments in the history of mathematics, in particular the mathematics that forms the core of the curriculum at the TU Delft; 2. the student is able to study, independently, the historical aspects of a mathematical subject or area and report on this study; 3. the student is able to study a historical subject or paper and report on this orally and in writing.	
Education Method	A mix of lectures, discussions, presentations by students and written work (an essay on a subject of the student's choice). Major emphasis is on active participation. Every student will give at least two oral presentations on some subject (prepared in advance). Attendance is compulsory.	
Literature and Study Materials	There is no specific text that will be used. Throughout the course we create a reader by offering reading material on Brightspace. Thanks to world-wide digitization efforts we can use many original texts. Some books can be used as background material. There will list of recommended books on Brightspace in due course.	
Assessment	There is no written exam. The grade will be based on short quizzes about the reading material, on the oral presentations, and on the essay. These will get their own grades that are weighed as follows: essay 40%, presentations 40%, quizzes 20%. The number of participants may necessitate some deviation from the rules. If that happens then this will be made known via Brightspace. For the whole course attendance is compulsory. The precise regulations will be available on Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	BSC AM students can choose either AM2520-H or AM2520-P. It is not allowed to enter both courses in the individual study programme.	
Co-Instructor	Prof.dr. J.M.A.M. van Neerven	

AM2520-P	Philosophy of Mathematics	6
Responsible Instructor	Prof.dr. J.M.A.M. van Neerven	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Specific prior knowledge, other than that covered in the first three semesters of the TW study, is not required. A broad interest in mathematics, science and philosophy in general and an open attitude are, however, required.	
Course Contents	In this course we investigate what mathematics is and how it relates to other (exact) fields of science. Central is "the applicability of mathematics as a philosophical problem": how can mathematics, as a formal language, have applications in the world around us? The following topics are covered: - what is knowledge? - what is science? - what is math? - foundations of mathematics: logic, truth, proof - main currents in the philosophy of mathematics - Wigner's "unreasonable effectiveness of mathematics" - the mathematical description of external reality: philosophical aspects of space, time and matter in classical mechanics, relativity theory, quantum mechanics and cosmology - mathematical aspects of determinism and causality - mathematical aspects of free will and self-awareness All these topics will be dealt with in their historical context. We will not only discuss current views and debates, but also the evolution of thinking on these subjects over the centuries, starting in Antiquity.	
Study Goals	After the course: 1. the student is familiar with the main points of the philosophy of mathematics and its applications in the natural sciences; 3. the student is able to independently study the philosophical aspects of a mathematical subject or sub-field and to report on this; 4. the student is able to explore and present a philosophical subject or article orally and in writing.	
Education Method	A mix of lectures, discussions, student presentations and written work (essay on one topic but choice). Great emphasis is placed on active student participation. Every student gives at least two oral presentations on a pre-prepared topic. Attendance is compulsory for this entire course. Depending on the situation surrounding COVID-19, the above can be adjusted.	
Literature and Study Materials	There is no course book. A reader is built up by offering weekly reading material via Brightspace. In the background we use some books, the titles of which will be announced in due time.	
Assessment	There is no written exam. Depending on the number of participants, assessment takes place on the basis of short tests, oral presentations, and/or a written essay. Separate grades are given for these, and the final grade is a weighted average of these. The details will be set out in the study manual that will be offered via Brightspace. If the final grade is 5.7 or less, a repair option will be offered. Attendance is compulsory for this entire course. The exact regulation can be found in the course description on Brightspace.	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) 1) Only one of the two courses AM2520-H ofwel AM2520-P can be attended for ECTS credit points, but not both. 2) A cap of 20 students applies. If enrollments exceed this number, students will be selected on the basis of their motivation.	
Co-Instructor	Dr. K.P. Hart	

AM2550	Advanced Statistics	6
Responsible Instructor	Dr. N. Parolya	
Contact Hours / Week x/x/x/x	0/0/2/0 hc; 0/0/2/0 wc; 0/0/4/0 prac	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Introduction to Statistics TW2080; Introduction to Probability Theory TW1080; Linear Algebra TW1030, TW2011	
Course Contents	Mathematical foundations of statistical methods and models used for multivariate data analysis with linear models. Application-oriented component for analyzing and interpreting the data with "R". Short outline of the topics: 1. Multiple linear regression (OLS and GLS) 2. ANOVA (one-way and two-way) 3. Generalized Linear Models (logit and probit) 4. Generalized Least Squares (heteroscedasticity and white errors)	
Study Goals	After successful completion of this course, the student is able to LO1: describe the definition of the multiple linear regression model; LO2: create a multivariate linear model, which fits the best the underlying data set; a: identify the relevant explanatory variables; b: find the least squares estimator; c: interpret the estimated coefficients; d: provide the corresponding confidence and prediction intervals; LO3: analyze the ANOVA model; a: recognize ANOVA from a general class of linear models; b: estimate its unknown parameters; c: provide the valid statistical inference of the model; d: interpret the final results in terms of mean profiles; LO4: analyze the Generalized Linear Model (GLM); a: define the GLM; b: estimate the GLM; c: provide a valid asymptotic inference; d: interpret the obtained results in terms of odds; LO5: handle the general linear models with heteroscedasticity and autocorrelation; a: verify whether the classical assumption of homoscedasticity is valid; b: apply weighted least squares method efficiently; c: interpret the white standard errors; LO06: give proofs of the main theoretical statements about the statistical properties of the estimators. LO07: apply the learned methods in practice using the statistical software R. (LO stays for learning objective)	
Education Method	Lectures, tutorials, computer sessions	
Literature and Study Materials	Syllabus via Brightspace, notes on Brightspace and from the lecture	
Books	Applied Regression Analysis and Generalized Linear Models by John Fox, SAGE Publications 2016 Stock, J. H. and M. W. Watson (2012). Introduction to Econometrics, Pearson.	
Assessment	Written or oral (in case the number of students <10) examination (70% of the final grade). Resit as written or oral examination (70% of the final grade), again depending on the number of students. Weekly compulsory computer exercises (30% of the final mark). In case of an insufficient final result, repair options may exist for the exam or computer assignments in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	A graphical calculator A sheet of A4 paper, handwritten on both sides.	
Judgement	Computer assignments: 5 best of 6 assignments amount to a grade: C Exam: grade T. Final grade= $0.3 \cdot C + 0.7 \cdot T$ Resit: grade R. Final grade for resit= $0.3 \cdot C + 0.7 \cdot R$	

AM2560	Applied Algebra: Codes	6
Responsible Instructor	Dr. J.A.M. de Groot	
Contact Hours / Week x/x/x/x	0/0/4/0 hc 0/0/4/0 wc	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch (on request English)	
Expected prior knowledge	Linear Algebra 1, Algebra 1	
Course Contents	The goal of this course is to show some practical applications of algebra. The course will have the following structure: -Introduction to coding theory (hamming distance, linear codes, parity check matrices and generator matrices, syndrome decoding) -An introduction to finite fields -Continuation of coding theory (cyclic codes, BCH codes, Reed Solomon codes) Due to a change in the number of teaching hours there is no time left for the short introduction to cryptography (classic, modern). The dutch version of the course contents is much much more extended.	
Study Goals	Understanding and being able to apply the materials and techniques of this course. Furthermore, the ability to apply experience and knowledge to standard as well as non-standard problems.	
Education Method	Four hours lecture and four hours exercise class per week. A lot of material (for example the lecture notes) is in dutch and will not be translated to english when the course is given.	
Literature and Study Materials	Coding Theory and Cryptography, the essentials, 2nd edition, written by D.R. Hankerson, D.G. Hoffman, and five others (published by CRC Press). Lecture slides (Brightspace) Extra exercises (Brightspace) An additional reader containing some extra theory and the extra extra exercises.	
Assessment	A three hour, written, final exam at the end of the third quarter, with a resit exam in the fourth quarter. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Judgement	A three hour, written, final exam at the end of the third quarter, with a resit exam in the fourth quarter.	
Co-Instructor	Dr.ir. J.H. Weber	

AM2570	Markov Processes	6
Responsible Instructor	Dr. C. Kraaikamp	
Contact Hours / Week x/x/x/x	0/0/6/0 hc; 0/0/2/0 instr	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The expected prior knowledge is the content of the first year course "Introduction to Probability Theory" (AM1080). Active knowledge of the students of the two partition theorems, of probability and moment generating functions and of conditional probabilities and conditional expectations is assumed.	
Course Contents	This is an introductory course on Markov chains, where time-discrete and continuous time Markov chains will be introduced, and their most fundamental basic properties studied. Properties as the matrix of transition probabilities (and in the continuous case the Q-matrix), stationary distribution, ergodicity of the Markov chain, recurrence and transience of states will be discussed. Very many examples will be presented to illustrate these notions.	
Study Goals	After this course you can work with the basic properties of time discrete and continuous time Markov chains. You can determine the matrix of transition probabilities (and in the continuous case also the Q-matrix), you can determine the invariant measure in the instances it exists and derive from this conclusion about the system you are studying, in particular the long-term behavior of this system. You should be able to use the theorems discussed in class effectively to solve the problems, and the various notions introduced in class (such as irreducibility and (a)periodicity) will be a part of your tool set.	
Education Method	This course will be taught in a standard classroom setting, with exercises hours. You have to prepare in advance for the exercise hours, and then during these exercise hour the exercise will be solved by you and the lecturer in an interactive way.	
Literature and Study Materials	In this course the same book (by Grimmett and Welsh) will be used as in AM1080. There is also a reader, which will be made available on the Brightspace page of this course. Other literature mentioned in class can always be found on-line in the library of the TU Delft, and is available for the students.	
Assessment	There will be weekly assignments, but the final grade is determined only by the written exam. The weekly assignments can be handed in for comments, but no grade will be given. The resit consists of a single written exam that counts for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Prof.dr. F.H.J. Redig	

AM2580	Mathematical Models in Biology	6
Responsible Instructor	Dipl.-Math. H. Yoldas	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course introduces prominent mathematical models and modeling approaches in biology ranging from population dynamics, cell signaling, reaction kinetics, epidemiology, and neuroscience. We will cover a variety of discrete (agent-based models, difference equations) and continuum (ordinary and partial differential equations (ODEs, PDEs)) mathematical models. We will employ diverse mathematical techniques including dimensional analysis, non-dimensionalization, stability analysis, parameter estimation, and numerical analysis to study these models. A special attention will be paid to the biological interpretation of the models (modeling parameters and the relevance of the modeling assumptions), their applicability to the real world and the implications of the mathematical results. Several computational tools will be introduced and applied during the lab hours. The lectures together with accompanying instructions and lab work will provide a profound introduction to mathematical biology.	
Study Goals	After successfully completing the course, you should be able to: LO1. Give a biological interpretation to a variety of discrete/continuum models. LO2. Classify mathematical models correctly, e.g., distinguish deterministic, stochastic; discrete, continuum; microscopic, mesoscopic, macroscopic; ODE, PDE models from one another. LO3. Identify and apply appropriate mathematical techniques to analyze a mathematical model. LO4. Interpret the results of the mathematical analysis biologically.	
Education Method	Lectures, instructions, assignments and computer labs.	
Computer Use	Your favourite software package (Python; Matlab, cell-based modeling software packages etc.)	
Literature and Study Materials	Mathematical Models in Biology by Leah Edelstein-Keshet	
Books	Mathematical Biology by James D. Murray	
Assessment	The final grade of the course consists of the following components: - Assignments (30%) - Written exam (70% (if the assignments are submitted) or 100 % (if no assignments is submitted)) Final grade calculation: $(0.3 * \text{Grade obtained from the assignments (if the total grade obtained from assignments is more than the written exam grade)} + 0.7 * \text{Written exam grade})$ If the assignments were not submitted or the written exam grade is higher than the one obtained from assignments, the written exam will weigh 100%. The resit consists of a single written exam that counts for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Pen and paper	
Co-Instructor	Dr.ir. E.G. Rens	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Applied Mathematics

3e jaar BSc TW 2023

Program Coordinator	Bachelorcoordinator: Dr. J.A.M. de Groot J.A.M.deGroot@tudelft.nl
Prerequisites	Be aware! The entry requirements have been recorded in the OER. If there is a difference between what is listed here and what had been recorded in the OER, the content of the OER is leading. To participate in the course AM3000 Bachelorproject, it is mandatory to have finished all courses of the first year and at least 40 EC of the second year and the second semester of the third year.
Introduction 1	Enrollment: For the (partial-)exams of all courses, you have to register via Brightspace/OSIRIS ultimately one week before the day of the exam. For more information, look at https://www.tudelft.nl/en/student/education/examinations/. Do you not want to participate in a study component for which you have enrolled? Please unroll yourself via Brightspace/OSIRIS!
Introduction 2	In the third year you will do a minor. For more information, see http://minors.tudelft.nl. If want to do a master after your bachelor, the now is the time to inform yourself. For this pay attention to the announcements on Brightspace about the Master Events!

AM3000	Bachelor Project	18
Responsible Instructor	Dr. J.G. Spandaw	
Responsible Instructor	Dr.ir. J.T. van Essen	
Responsible Instructor	Dr. T.W.C. Vroegrijk	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3 4 Different, to be announced	
Start Education	3	
Exam Period	none	
Course Language	Dutch	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Applied Mathematics

Minor (30 EC)	
Director of Education	Bachelorcoordinator: Dr. J.A.M. de Groot J.A.M.deGroot@tudelft.nl

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Applied Mathematics

keuzevakken 3e jaar BSc TW 2023	
Director of Education	Bachelorcoordinator: Dr. J.A.M. de Groot J.A.M.deGroot@tudelft.nl

AM3500	Mathematics Seminar	6
Responsible Instructor	Dr. M.V. Gnann	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Course Contents	This course has a different topic every year. For the academic year 2023/2024 this course is organized by the Mathematical Physics group and the topic is calculus of variations. A collection of text books and other reading material will be made available to students, from which the talks for this course can be prepared. The calculus of variations is a topic in mathematical physics and mathematical analysis that has a rich history, yet is still very active thanks in part to its many applications in classical and quantum mechanics, micromagnetics, shape memory alloys, elasticity theory, optimization, differential geometry, machine learning, etc. We will deal mostly with the one-dimensional version of the theory, but will also deal with some of the higher-dimensional theory. Topics may include, but are not limited to 1D calculus of variations: * Euler-Lagrange Equation * Second Variation * Minimal Surfaces * Dido's Problem * Brachistochrone Problem * Corner Conditions * Isoperimetric Constraints * (Non-)Holonomic Constraints * Noether's Theorem * Two-Body Problem Multidimensional calculus of variations * The Direct Method of the Calculus of Variations * Young Measures * Gamma Convergence * Gradient Flows	
Study Goals	See the Brightspace page of this course.	
Education Method	The course mainly consists of lectures delivered by the students. The role of the instructors is to coach the students, provide feedback on the lectures, as well as to give additional perspectives on the course material. Necessary prerequisites beyond the courses from the first two years may be taught by the instructors depending on the background of the students. The course material is divided among the students in the beginning of the course. The students prepare lectures about the assigned material including detailed proofs of relevant results.	
Literature and Study Materials	The course treats a selection of chapters from the following books (all available as e-book or hardcopy via the library) - Kielhöfer, Hansjörg Calculus of variations. An introduction to the one-dimensional theory with examples and exercises. Translated from the 2010 German original. Texts in Applied Mathematics, 67. Springer, Cham, 2018. xvi+227 pp. ISBN: 978-3-319-71122-5; 978-3-319-71123-2 49-01 - Jost, Jürgen; Li-Jost, Xianqing Calculus of variations. Cambridge Studies in Advanced Mathematics, 64. Cambridge University Press, Cambridge, 1998. xvi+323 pp. ISBN: 0-521-64203-5 - Rindler, Filip (4-WARW-MI) Calculus of variations. Universitext. Springer, Cham, 2018. xii+444 pp. ISBN: 978-3-319-77636-1; 978-3-319-77637-8 49-01 - Dacorogna, Bernard (CH-LSNP) Direct methods in the calculus of variations. Applied Mathematical Sciences, 78. Springer-Verlag, Berlin, 1989. x+308 pp. ISBN: 3-540-50491-5 49-02 (49A50 73C50 73C60 73K40)	
Prerequisites	Additional material may be provided during the course. Year 1: Mathematical Structures (AM1010), Analysis 1 and 2 (AM1040 and AM1070) Year 2: At least two out of the following courses: Ordinary Differential Equations (AM2030), Partial Differential Equations (AM2070), and Real Analysis (AM2090)	
Assessment	The students' lectures are graded (70% of the final grade) and the course is concluded with an oral exam (30% of the final grade). In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Tags	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Intensive Mathematics	
maximum aantal deelnemers	Due the special nature of the course (student lectures) the maximum number of participants is limited to 12. If the number of enrollments exceeds this number, priority will be given to the students from the Excellence Program. Secondary consideration will be given to performance on the prerequisite courses. Please enroll before the start of the course on the Brightspace page (once it is available) if you are planning to participate in this course. Enrollment after the first meeting is no longer possible.	
Co-Instructor	Dr. Y. van Gennip	

AM3510	Mathematical Physical Models	6
Responsible Instructor	Dr. B.J. Meulenbroek	
Contact Hours / Week x/x/x/x	4/4/0/0 hc	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Ordinary Differential Equations AM2030, a first course in partial differential equations in which separation of variables is treated	
Course Contents	In this course we will develop a number of mathematical models for physical phenomena, especially in the context of continuum mechanics. All the models discussed are partial differential equations. The focus of the course is not on how to solve these equations, but how to derive them from the underlying physical laws, such as mass and energy conservation.	
Study Goals	The student is able to: 1. Formulate conservation laws 2. Use these laws to derive differential equations 3. Construct mathematical models for a number of physical phenomena related to continuum mechanics, such as displacement of material under loads, travelling of sound waves through a medium,.	
Education Method	Lectures. The first series of lectures are given by dr. B. Meulenbroek, the second series of lectures by dr. J.L.A. Dubbeldam	
Assessment	Homework exercises and an oral exam. The final mark for the course is the mark of the exam. Accordingly, one may obtain additional bonus points based on the quality of the assignment. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Judgement	Homework exercises 50% and oral exam 50% of the final grade	
Co-Instructor	Dr. J.L.A. Dubbeldam	

AM3530	Numerical Methods 2	6
Responsible Instructor	M.E. Kootte	
Contact Hours / Week x/x/x/x	0/0/4/0 hc; 0/0/4/0 prac/wc	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Numerical Analysis 1 (AM2060), Partial Differential Equations (AM2070)	
Course Contents	Finite differences, finite volumes in $\mathbb{R}^2$, error analysis, discrete and continuous maximum principles, processing of boundary conditions, time integration.	
Study Goals	The focus of this course is on two out of three most-common discretisation techniques: the finite difference method and the finite volume method. Besides the introduction, derivation and application of these methods, we start with the theoretical principles of differential equations that are required to analyse PDEs before we can discretise them. Then, we introduce these two discretisation techniques. Next to that, we will introduce iterative methods to solve non-linear PDEs and lastly we will see several numerical time integration methods, that we will introduce by means of the heat equation and the wave equation. More concretely, at the end of the course, you should be able to: - Understand analytical properties of partial differential equations (PDEs) and the role of boundary and initial conditions. You should be able to classify these PDEs and investigate mathematical properties that are important to obtain solution systems of PDEs: well-posedness, compatibility, analytic maximum principle - Explain the finite difference method (FDM) and apply it to discretise 1D and 2D linear systems of equations. You should be able to analyse the accuracy of the method and understand how the discrete maximum principle relates to the analytic maximum principle. - Explain the finite volume method (FVM) and apply it to discretise 1D and 2D linear systems of equations. You should understand how the FVM is applied to PDEs in conservative form. - Apply two techniques to obtain solutions of non-linear PDEs, namely: the fixed-point scheme and Newtons method. You should be able to motivate and implement sensible initial estimates and analyse its convergence properties. - Construct and implement numerical schemes for time-dependent PDE's, which will be introduced to you by means of two time-dependent equations: The diffusion and the wave equation. Furthermore, you should be able to design and analyse these numerical schemes on convergence, stability, and consistency and understand what dispersion and dissipation means.	
Education Method	Lectures, instructions + project	
Books	J. van Kan, A. Segal, F. Vermolen and H. Kraaijevanger. Numerical Methods for Partial Differential Equations, Delft Academic Press, Delft, 2019.	
Assessment	Project + oral exam (oral exam = 1 hour written exercises + 0.5 hour oral discussion) The final mark for the course is the mark for the exam. Accordingly, one may obtain additional bonus points based on the quality of the assignment. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	This course can not be taken in combination with TW3730TU Numerical Methods for Differential Equations	
Judgement	0.2*Project report+0.8*Oral exam	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

AM3540	Inverse Problems	6
Responsible Instructor	Dr. H.N. Kekkonen	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic linear algebra and matrix calculus. Familiarity with Matlab is useful but not mandatory.	
Course Contents	The goal of inverse problems is to determine the cause from the effect. For example, the cause can be a sharp image. The effect is a blurred and noisy image given as a result of a camera that is badly focused. Recovering the sharp image from the noisy measurement is a classic example of an inverse problem. Mathematically we can model blur in an image by convolution. The observed image is interpreted as a sharp image convolved with a kernel and corrupted with random noise. The related inverse problem of recovering a sharp image is called deconvolution. This inverse problem is ill-posed, meaning that the solution is highly sensitive to modelling errors and measurement noise. A stable solution can be produced using regularisation. Topics: Naïve inversion, (discrete) convolution, truncated SVD, Tikhonov regularisation, and if time permits Bayesian approach.	
Study Goals	After this course you should - Know how to model a measurement process as a matrix equation and detect if it leads to an ill-posed inverse problem. - Understand why ill-posed problems require regularisation. - Be able to design and implement a reconstruction method to recover the unknown object of interest from indirect and noisy measurements. - Be able to solve discrete inverse problems using Matlab.	
Education Method	Lectures and exercise +computer labs	
Assessment	80% Final exam and 20% homework assignments. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr. Y. van Gennip	

AM3550	Graph Theory	6
Responsible Instructor	Dr. A. Bishnoi	
Contact Hours / Week x/x/x/x	0/0/8/0 colstr	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Linear algebra	
Course Contents	Graph theory is the main subject of this course. We will study a number of central topics from graph theory, including: 1. paths, cycles and trees 2. bipartite graphs 3. matching theory 4. graph colouring 5. planar graphs 6. spectral graph theory Our primary focus will be on theory and proofs, but throughout the course we will keep describing some important real-life applications of the concepts introduced.	
Study Goals	By the end of this course, you will be able to: 1. Express yourself fluently in the language of graph theory. 2. Apply various proof techniques like induction, double counting, and extremality arguments, to graph theory problems. 3. Explain the proofs of some of the central theorems from graph theory. 4. Solve combinatorial problems using the theory developed in the course.	
Education Method	Lectorials	
Books	Book: Graph Theory, by R. Diestel Book: Introduction to Graph Theory, D. West	
Assessment	Written examination The resit consists of a single written exam that counts for 100%.	
Permitted Materials during Tests	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) None	
Judgement	The exam grade is the final grade	
Co-Instructor	Dr. F.M. de Oliveira Filho	

AM3560	Advanced Probability	6
Responsible Instructor	J. Komjáthy	
Contact Hours / Week x/x/x/x	0/0/8/0 colstr	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The goal of the course is to extend the basic notions of probability with an analytic toolkit. Rigorous probability is closely related to real and complex analysis. We will explore these connections and learn how concepts from real analysis translate to probability and vice versa. In particular we will construct a general probability space and perform computations of expectations, and conditional expectation of random variables therein. We will treat different types of convergences of random variables, Borel-Cantelli Lemmas, Kolmogorov's 0-1 law. We will learn various analytic methods to prove convergence of random variables: characteristic functions, Fourier inversion, generating function. Laws of Large Numbers and general Central Limit Theorems will be discussed. If time permits, we will learn about stable distributions.	
Study Goals	At the end of the course, you are able to construct a general probability space. You are able to recognize, analyse, and construct (abstract) random variables as measurable functions, and sigma algebras. You are able to define expectation and conditional expectation of arbitrary random variables, even when the conditioning is a 0 probability event and the random variable does not have a density. You are able to link the abstract definition of conditional expectation to earlier courses, e.g. conditional expectation in the setting of two dimensional random variables, and compute them using the combination of abstract theory and integration. You are able to compute the characteristic functions of well-known random variables, and use the characteristic function method to prove distributional identities. You are able to apply the tower rule to simplify long integrations into easy-to-compute formulas using conditional expectation. (which will be very useful in Stochastic Processes later on). You are able to invert the characteristic function of discrete random variables. You are able to apply Markov's and Chebyshev's inequality to bound the expectation and variance of sums of random variables. You are able to prove the Central Limit Theorem using characteristic function. You are able to prove Central Limit Theorem for non-iid random variables, you can apply and check Lindeberg's condition. Given a sequence of random variables, you are able to tell whether they converge almost surely, in probability, in L^1 or in distribution. You are able to apply the Borel-Cantelli lemmas to prove that a given a sequence of random variables converges almost surely. You are able to combine analysis (of characteristic functions) to prove that a given a sequence of random variables converges in distribution. You are able to prove the Weak Law of Large Numbers using Chebyshev's inequality. You remember the limitations of the Law of Large Numbers, and recognize when it does not hold for a sequence of random variables. You are able to estimate the maximum of a collection of heavy tailed random variables using Markov's inequality and elementary computations.	
Education Method	The course content is based on the book "Probability Essentials" by J. Jacod and P. Protter. There will be 4h lectures and 2h exercises and 2h discussion of the exercises per week with homework (twice).	
Assessment	The assessment will consists of two homework assignments and one final written exam. One needs at least 5.0 in the assignments to be allowed to sit at the exam. The assignments do not count towards the resit. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	There is no additional material allowed during the exam. For the home assignments there is no restriction on the material allowed.	
Judgement	The grade will be calculated as follows: - 40% of the grade is based on home assignment (two assignments throughout the course); - 60% of the grade is based on the final written exam. The final exam must have a grade larger or equal to 5 to pass the course.	
Co-Instructor	Prof.dr. F.H.J. Redig	

AM3570	Fourier Analysis	6
Responsible Instructor	F. Bartolucci	
Contact Hours / Week x/x/x/x	4/4/0/0 colstr	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	- Real Analysis AM2090 - Complex Function Theory AM2040	
Course Contents	The aim of the course is to provide an introduction to Fourier analysis, on the torus and on the real line. The course will cover the following subjects: Fourier series: - Fourier series of periodic absolute integrable functions - Cesàro and Abel summability - Fourier series of periodic square integrable functions - pointwise convergence - applications to PDEs - finite Fourier series (optional). The Fourier transform on the real line: - Fourier transform of absolute integrable functions - convolutions and approximate identities - Fourier inversion theorem - Fourier transform of square integrable functions - Poisson summation formula - Paley-Wiener theorem - Shannon theorem (optional). The Fourier transform of tempered distributions.	
Study Goals	At the end of the course, you should be able to: - reproduce the notions and the statements of the theorems covered in the course - reproduce the proofs of the results covered in the course in a clear and rigorous way - apply theoretical results and proof techniques in solving new exercises.	
Education Method	The course consists of frontal lectures where the theory is explained and where applications are discussed. These are integrated with exercise classes.	
Literature and Study Materials	Lecture notes are available on Brightspace.	
Assessment	The exam consists in an oral exam.	
Co-Instructor	Prof.dr. J.M.A.M. van Neerven	

AM3580	Differential Geometry	6
Responsible Instructor	Dr. B. Janssens	
Contact Hours / Week x/x/x/x	0/0/4/0 hc; 0/0/4/0 wc	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Linear algebra I and II (TW1030 and AM2010), and Analysis I and II (AM1040, AM1070). It is an advantage if you have taken a course on topology (e.g. AM3590), but otherwise it is still possible to follow this course with a little extra effort. The necessary material on topology is gathered in appendix A of the course notes, I recommend that you read at least A1 and A2 (2 pages) before the start of the course.	
Course Contents	Differential geometry is the study of smooth manifolds, such as spheres, tori, hyperbolae, etc. Since these spaces behave locally like a vector space, they are amenable to techniques from analysis and linear algebra. The focus of this course will be on Riemannian geometry, the study of metric spaces in a smooth context. At the end of the course, we will discuss the link between (pseudo)-Riemannian geometry and Einstein's theory of general relativity.	
Study Goals	At the end of this course, you should able to: 1) explain the basic notions of differential geometry, such as smooth manifolds, tangent bundles, Riemannian metrics, and geodesic paths. 2) calculate the Riemann curvature tensor for a given Riemannian metric, and apply this in the context of connections, curvature and geodesics. 3) explain and apply a number of basic theorems of differential geometry, such as Gauss' Theorema Egregium. 4) interpret and explain Einstein's equation for the gravitational field.	
Education Method	Lectures and exercise sessions	
Literature and Study Materials	Lecture notes, to appear on Brightspace.	
Assessment	Written examination and resit The resit consists of a single written exam that counts for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

AM3590	Topology	6
Responsible Instructor	Dr. K.P. Hart	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Mathematical Structures, Analysis 1 and 2, Real Analysis, and Algebra 1.	
Summary	Disclaimer: the study guide information may change depending on the developments around the coronavirus. Only on September 1st will it be in its final form.	
Course Contents	The course is divided into two parts: - the topology of Euclidean spaces: a proof that the different powers of the real line are topologically distinct by formulating an invariant, topological dimension, that assigns the natural number n to the n-th power of $\mathbb{R}$. An important intermediate result will be Brouwer's Fixed-point Theorem. - An introduction to General Topology: the definition of a topological space and various elementary properties; a short treatment of the basic separation axioms, including Hausdorffness (complete) regularity and normality; existence of continuous functions; the results of Urysohn and Tietze; compactness, its uses, the general form of Tychonoff's theorem and results of the Arzela-Ascoli type that characterize compactness in function spaces. If students express the desire to explore some other topic in depth then that is up for consideration.	
Study Goals	At the end of the course - the student recognizes the need to distinguish the various finite-dimensional Euclidean spaces - the student is able to explain how this topological difference can be determined and in particular how the dimension can be characterized topologically and the role of Brouwer's fixed-point theorem in this - the student can explain what general topological spaces are and illustrate this using various (well-known) examples described during the course - the student knows the most important separation properties and can describe their mutual relationships - the student can describe the role of compactness in general topological spaces - the student knows the properties of compactness that are important in (Functional) Analysis, such as Tychonoff's theorem and the theorem of Arzela-Ascoli	
Education Method	Lectures and possibly some exercise classes. There will be plenty of (homework) exercises to help absorb the topological notions.	
Literature and Study Materials	There will be lecture notes.	
Assessment	This will depend on the number of students: a written exam with a large population, an oral exam with a small population. The cut-off will be at 15 students. The resit consists of an exam that counts for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr. B. Janssens	

F. Bartolucci

Department	Analysis
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Dr. A. Bishnoi

Department	Discrete Wiskunde en Optimalisering
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